NIFTY-50 AI Investment Intelligence Platform

A Risk-Aware, Explainable Decision-Support System for Quantitative Asset Allocation

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Executive Summary

In quantitative finance, forecasting equity returns is highly challenging due to the low signal-to-noise ratio of daily market data. The **NIFTY-50 AI Investment Intelligence Platform** is an end-to-end quantitative decision-support system that ingests historical National Stock Exchange (NSE) data from January 2000 to April 2021 to generate mid-term return forecasts, construct risk-constrained portfolios, and provide transparent explanations for investors.

The platform integrates four core components: a **Stock Predictor Engine**, a **Portfolio Construction Module**, a **Risk Assessment Module**, and an **Explainable AI (XAI) Framework**, all deployed via an interactive **Streamlit Dashboard**. The forecasting engine implements Ridge Regression and LightGBM models validated using a purged walk-forward cross-validation strategy. Recognizing the limits of short-term forecasts in efficient markets—where out-of-sample directional prediction accuracy is 49.6% for LightGBM—the platform focuses on downstream portfolio optimization. It utilizes Ledoit-Wolf covariance shrinkage, Modern Portfolio Theory (MPT), and Black-Litterman models to blend machine learning views with market equilibrium weights.

Historical backtests spanning January 2010 to December 2018 demonstrate that the platform improves portfolio stability. The Aggressive (Black-Litterman) profile achieved an annualized return (CAGR) of 7.73%, annualized volatility of 13.25%, and a maximum drawdown of -23.63%, compared to a benchmark drawdown of -28.01%, demonstrating the effectiveness of diversification, risk-aware portfolio construction, and drawdown mitigation under realistic market conditions.

1. Introduction

Indian equity markets, represented by the benchmark NIFTY 50 index, exhibit complex price dynamics, non-linear dependencies, and time-varying volatility regimes. For institutional asset managers, generating consistent risk-adjusted returns across index constituents is challenging due to the stochastic nature of daily market returns.

Rather than relying on stock-price predictions in isolation, this platform translates noisy market signals into diversified portfolio allocations. Quantitative asset management face several key obstacles:

1. **Low Signal-to-Noise Ratio**: Daily price movements are highly stochastic, making models trained on raw prices prone to overfitting.
2. **Lookahead Bias & Data Leakage**: Standard cross-validation fails on financial time series because overlapping return labels leak future information into training sets.
3. **Estimation Error**: Classical Mean-Variance Optimization (MVO) is highly sensitive to return assumptions, often generating concentrated, unstable portfolios out-of-sample.

4. **Model Opacity**: Machine learning models lack interpretability, which limits investor trust during market regime shifts.

The NIFTY-50 AI Investment Intelligence Platform addresses these issues through:

- Engineering momentum, volatility, and systematic features with zero lookahead bias.
- Implementing a purged walk-forward validation framework to prevent data leakage.
- Utilizing covariance shrinkage and Black-Litterman models to build diversified portfolios.
- Integrating SHAP values to explain individual stock signals and model drivers.
- Deploying these modules in an interactive Streamlit dashboard.

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2. Dataset & Exploratory Analysis

The platform relies on historical market data and metadata provided within the project workspace, spanning the period from **January 3, 2000, to April 30, 2021**.

2.1 Data Ingestion & Schema

The ingestion pipeline integrates five primary data sources:

1. **Constituent Equity Data (`data/nifty50/`)**: Daily records of individual NIFTY 50 equities (e.g., `INFY.csv`, `TCS.csv`). Key fields include `Open`, `High`, `Low`, `Close`, `Volume`, `Turnover` (rupee value traded), and `VWAP` (Volume-Weighted Average Price).
2. **Benchmark Index (`data/archive (1)/Datasets/INDEX/NIFTY 50.csv`)**: Daily records of the NIFTY 50 index, serving as the master market index and investment benchmark.
3. **Volatility Index (`data/archive (1)/Datasets/INDEX/INDIA VIX.csv`)**: Option-implied market volatility close prices (available from July 2010 onwards), reflecting market fear and risk regimes.
4. **Sectoral Indices**: Sector-specific indexes (e.g., `NIFTY BANK.csv`, `NIFTY IT.csv`) mapped to individual stock sectors to measure sector-relative momentum.
5. **Metadata (`stock_metadata.csv`)**: Mapping of tickers to their corresponding industries (e.g., IT, Financial Services, Consumer Goods).

Data Schema Table

Field	Source File	Data Type	Description / Preprocessing
Date	Stock & Index CSVs	`datetime64[ns]`	Canonical index timeline; sorted chronologically.
Close	Stock & Index CSVs	`float64`	Daily closing price used for return calculations.
Open	Stock CSVs	`float64`	Used for executing portfolio fills at $t+1$.
Volume / Turnover	Stock CSVs	`float64`	Traded quantity and value; used for liquidity features.

Field	Source File	Data Type	Description / Preprocessing
VWAP	Stock CSVs	`float64`	Volume-weighted average price; proxy for clean entry.
Industry	`stock_metadata.csv`	`object` (string)	Categorical classification for sector caps.
VIX Close	`INDIA VIX.csv`	`float64`	Implied volatility level; missing values pre-2010.

2.2 Exploratory Data Analysis (EDA)

Exploratory analysis of the NIFTY-50 dataset highlights several structural features of the Indian equity market:

- **Long-Term NIFTY Behavior and Growth**: The NIFTY 50 index level (2000–2021) shows long-term growth driven by economic expansion, punctuated by cyclical downturns. The index experienced a significant correction during the 2008 Global Financial Crisis (losing more than 50% from peak to trough), followed by recovery, consolidations in 2011 and 2015, and the high-stress liquidation of March 2020 during the COVID-19 outbreak. This long-term profile indicates that index-wide drawdowns are severe but temporary, highlighting the importance of drawdown mitigation during market corrections.
- **Sector Composition and Rotations**: Sector distributions in `stock_metadata.csv` reveal high weights in Financial Services, Information Technology (IT), Energy, and Consumer Goods (FMCG). Cyclical sectors (like Metals and Financials) display high betas during expansionary regimes but experience heavy drawdowns during crises. Conversely, defensive sectors (such as FMCG and Pharmaceuticals) show more stable returns during corrections. These dynamics confirm that sector-level caps are necessary to avoid concentration in highly cyclical industries.
- **Return Distributions and Volatility Clustering**: Consistent with stylized facts of financial assets, daily log returns of NIFTY 50 constituents display heavy tails, negative skewness, and high excess kurtosis. Volatility is not constant; it clusters over time, meaning that periods of high volatility (e.g., 2008, 2020) are grouped together. This clustering suggests that simple rolling variance models underrepresent tail risk, making metrics like Value at Risk (VaR) and Conditional Value at Risk (CVaR) essential for downside protection.
- **Correlation and Co-movements**: Asset returns show high positive correlation, which increases during market liquidations. For example, financial stocks (e.g., `HDFCBANK`, `ICICIBANK`, and `SBIN`) and IT stocks (e.g., `INFY` and `TCS`) exhibit strong intraday co-movements. During major market sell-offs, correlations tend to converge toward 1.0, reducing the diversification benefit of equity-only portfolios and highlighting the need for cash sleeves.
- **Volatility Regimes and INDIA VIX**: The India VIX index shows a strong negative correlation with the NIFTY 50 price level. High VIX regimes (>25%) indicate volatility spikes and market fear, during which technical indicators tend to decay. Low VIX regimes (<15%) represent stable trend-following conditions. The transition of VIX levels acts as a regime indicator for active risk management.

Summary of Key EDA Findings

Observation	Investment Implication
Volatility Clustering	Risk levels change through time, requiring adaptive covariance estimation.

Observation	Investment Implication
Correlation Spikes	Diversification benefits diminish during market stress, requiring cash sleeves.
Sector Leadership Rotates	Sector exposure constraints are necessary to prevent concentration risk.
Heavy-Tailed Returns	Standard deviation underrepresents tail risk, requiring VaR and CVaR.
India VIX Spikes	Coincide with market drawdowns, acting as macro risk filters.

3. Feature Engineering

The feature extractor `src/features.py` generates more than 50 technical, momentum, volatility, and market-relative indicators.

3.1 Mathematical Formulations of Key Features

1. **Wilder's Relative Strength Index (RSI)**: Measures momentum over a 14-day trailing window:

$$RSI_t = 100 - \frac{100}{1 + RS_t}, \quad RS_t = \frac{\text{EMA}_{14}(\max(\Delta P_t, 0))}{\text{EMA}_{14}(-\min(\Delta P_t, 0))}$$

2. **Systematic Beta (β_{60})**: Measures equity sensitivity to systematic market movements over a rolling 60-day window:

$$\beta_{60,t} = \frac{\text{Cov}_{60}(R_{\text{stock},t}, R_{\text{market},t})}{\text{Var}_{60}(R_{\text{market},t})}$$

3. **Volatility Measures (Annualized)**: Rolling standard deviations of daily returns, scaled to annual units:

$$\sigma_{\text{ann},w} = \text{Std}_w(R_{\text{stock},t}) \times \sqrt{252}$$

4. **Average True Range (ATR)**: Measures absolute price volatility over a trailing 14-day window:

$$ATR_{14,t} = \text{EMA}_{14}(\max(H_t - L_t, |H_t - C_{t-1}|, |L_t - C_{t-1}|))$$

5. **Sector Momentum**: Computed as the 21-day rate of change of the specific sector index:

$$\text{Sector Mom}_{21,t} = \frac{\text{Sector Close}_t}{\text{Sector Close}_{t-21}} - 1.0$$

3.2 Feature Categories Summary

Category	Key Indicators	Calculation Window	Description
Trend	`SMA_10`, `SMA_50`, `SMA_200`, `EMA_200`	10 to 200 Days	Captures support levels and long-term trend direction.

Category	Key Indicators	Calculation Window	Description
Momentum	`RSI_14`, `MACD_line`, `MACD_signal`, `ROC_21`	5 to 26 Days	Measures price velocity and overbought/oversold states.
Volatility	`ATR_14`, `Volatility_20`, Bollinger Bands	10 to 60 Days	Quantifies historical volatility and channel range limits.
Liquidity	`Volume_Momentum`, `Turnover`	20 Days	Identifies volume surges relative to trailing average.
Market/Sector	`Beta_60`, `Relative_Strength_21`, `Sector_Ret`	21 to 60 Days	Captures systematic risk and sector-relative outperformance.
Regime (VIX)	`VIX_level`, `VIX_ROC_5`	5 Days	Tracks macro-level implied volatility and trend dynamics.

4. Methodology

The platform connects statistical forecasting to risk-constrained optimization in a modular sequence.

4.1 Stock Predictor Engine

The prediction model targets **medium-term forward log returns over a 21-day holding horizon**:

$$y_t = \frac{\text{Close}_{t+21} - \text{Close}_t}{\text{Close}_t}$$

The model features are trained on a wide panel that is reshaped to a long format (Date, Symbol) using `src/models.py`.

4.1.1 Validation Methodology: Purged Walk-Forward Cross-Validation

Financial time series violate the independent and identically distributed (i.i.d.) assumption. Because the target is a 21-day forward return, targets at t and $t+1$ overlap. Using standard cross-validation leads to lookahead leakage.

Purged walk-forward cross-validation was used to eliminate look-ahead bias. Training employed a rolling 1,260-day window followed by a 21-day purge gap and a 252-day out-of-sample test window. The process was repeated using annual rolling steps.

This validation yields three distinct folds, spanning May 26, 2015, to June 11, 2018, encompassing 15,120 out-of-sample stock-day observations.

4.1.2 Model Implementations

We evaluate two model structures:

1. **Ridge Regression**: Linear baseline with L2 regularization ($\alpha = 1.0$) to control collinearity.

2. **LightGBM Regressor**: Non-linear tree-based ensemble designed to capture feature interactions (hyperparameters: `max_depth` = 4, `learning_rate` = 0.05, `n_estimators` = 500, `min_data_in_leaf` = 50, and `subsample` = 0.8).

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5. Portfolio Construction & Risk Management

Model predictions are not used directly to execute directional trades. Instead, predictions are passed to a portfolio construction module that filters out-of-sample estimates based on investor profiles.

```
AI Decision-Support Allocation Pipeline:
[Raw Market Data] ──> [Features] ──> [ML Predictor] ──> [Rank Filter] ──> [Shrunk
Covariance] ──> [Optimizer] ──> [Allocations]
```

5.1 Personalized Investment Profiles

The system implements three primary risk-profile models:

1. **Conservative Profile (Minimum Variance)**: Minimizes the portfolio's expected variance, ignoring return forecasts.

$$\min_w w^T \Sigma w, \quad \text{subject to} \sum w_i \leq 1.0,\ w_i \geq 0,\ w_i \leq 0.05$$

2. **Balanced Profile (Maximum Sharpe Ratio)**: Identifies the tangency portfolio that maximizes expected returns relative to portfolio risk:

$$\max_w \frac{w^T \mu - R_f}{\sqrt{w^T \Sigma w}}, \quad \text{subject to} \sum w_i = 1.0,\ w_i \geq 0,\ w_i \leq 0.10$$

3. **Aggressive Profile (Black-Litterman)**: Blends market equilibrium weights with the machine learning model's forecasts. The LightGBM predicted returns are converted into percentile ranks and mapped to active views. This stabilizes expected returns (μ_{BL}) and feeds them into a max-Sharpe optimizer:

$$\max_w \frac{w^T \mu_{BL} - R_f}{\sqrt{w^T \Sigma w}}, \quad \text{subject to} \sum w_i = 1.0,\ w_i \geq 0,\ w_i \leq 0.15$$

Constraints and Risk Boundaries

Profile	Primary Optimizer	Individual Stock Cap	Sector Exposure Cap	Cash Sleeve Policy
Conservative	Minimum Variance	5%	20%	Dynamic allocation to Cash when caps prevent full deployment.
Balanced	Maximum Sharpe	10%	25%	Fully invested in equity assets (sum of stock weights = 1.0).

Profile	Primary Optimizer	Individual Stock Cap	Sector Exposure Cap	Cash Sleeve Policy
Aggressive	Black-Litterman	15%	30%	Fully invested in equity assets; tilts toward active ML views.

5.2 Robust Fallback Hierarchy

During market crises, covariance matrices can become ill-conditioned. If the primary optimizer fails to converge, the platform executes a robust fallback hierarchy:

1. **Primary Optimizer** (e.g., Black-Litterman / Max Sharpe)
2. **Hierarchical Risk Parity (HRP)**: Computes allocation based on tree-based clustering of the covariance matrix.
3. **Equal-Weighted Portfolio**: Equal division of capital across the active universe.

For the **Conservative** profile, a dynamic cash allocation sleeve is implemented. If the 5% stock and 20% sector caps restrict the optimizer from allocating the entire capital to equities without violating covariance bounds, the residual capital is allocated to a CASH symbol, earning the daily risk-free rate.

5.3 Risk Assessment Module

The platform computes the following risk metrics:

- **Annualized Volatility**: Enforces annual scaling of variance.
- **Sharpe Ratio**: Ratio of excess return to volatility, using a risk-free rate of 6.5%.
- **Sortino Ratio**: Volatility replaced by downside semi-deviation.
- **Maximum Drawdown (MDD)**: Captures the largest peak-to-trough decline in cumulative value.
- **Historical Value at Risk (VaR)**: The 5th percentile of the historical daily returns distribution.
- **Historical Conditional Value at Risk (CVaR)**: The average loss in the worst 5% of cases.

To avoid the instability of sample covariance matrices, we implement **Ledoit-Wolf Covariance Shrinkage**. This method pulls the sample covariance matrix toward a constant-correlation target, minimizing estimation error and ensuring the matrix remains positive semi-definite (all eigenvalues ≥ 0).

5.4 Backtesting Framework

The backtesting engine `src/backtest.py` evaluates performance using a monthly rebalancing schedule over the historical window from **January 2010 to December 2018**.

- **Execution Timing and Open Fill Policy**: To prevent lookahead bias, signals are generated at the **Close of trading day t** (month-end) using features calculated up to Close(t) only. The trades are executed at the **Open of trading day $t+1$**. Portfolios are filled at the next session's Open price, not the closing price of the signal day. Same-day returns on the execution day are calculated as Open-to-Close, and subsequent days are calculated Close-to-Close.
- **Transaction Cost Modeling**: The engine enforces a **0.10% (10 bps) one-way transaction cost** on all equity rebalancing trades:

$$\text{Transaction Cost}_t = 0.0010 \times \sum_i |w_{i,t} - w_{i,t^-}|$$

This cost directly penalizes high-turnover strategies, ensuring that the backtest represents realistic net-of-cost returns. Transaction costs are not applied to cash transfers.

6. Explainable AI Framework

To build investor trust, the platform integrates model explainability using tree-based split frequencies and SHAP values.

6.1 SHAP Feature Attribution

SHAP (SHapley Additive exPlanations) values assign an attribution value to each feature, representing its contribution to the model's prediction relative to the base average prediction. The table below details the top 10 features by mean absolute SHAP value extracted from the repository output `outputs/shap_feature_importance.parquet`:

Rank	Feature Name	Mean Absolute SHAP Value	Mean SHAP Value (Direction)	Financial Interpretation
1	`EMA_200`	0.015786	+0.010546	Positive trend indicator; stocks above long-term EMA have higher expected returns.
2	`SMA_200`	0.011234	-0.005526	Long-term support/resistance indicator.
3	`SMA_50`	0.007575	-0.004194	Medium-term trend indicator; negative coefficient implies short-term mean reversion.
4	`Beta_60`	0.006286	-0.001618	High systematic risk is penalized during market consolidations.
5	`ATR_14`	0.006077	+0.002561	Higher absolute volatility correlates with higher risk premium returns.

7. Dashboard & Deployment

The platform is deployed as an interactive Streamlit dashboard. It organizes quantitative financial analytics into one landing page and four analytical dashboard modules to support investor decision-making.

Figures 1–4 illustrate the deployed Streamlit application and its core analytical modules. Figure 5 presents the model insights and SHAP-based global feature importances.

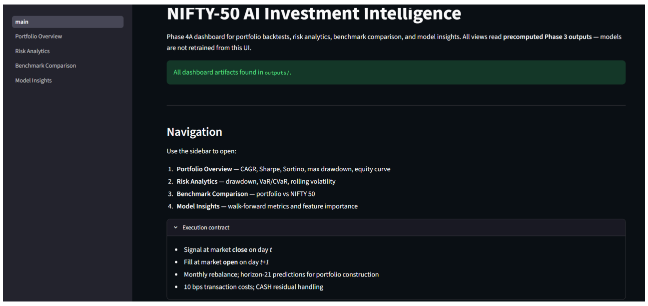

Figure 1: Home Dashboard Layout

Figure 2: Portfolio Overview and Equity Curve

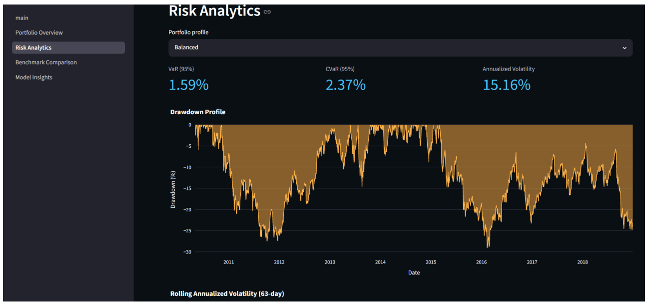

Figure 3: Risk Analytics Dashboard

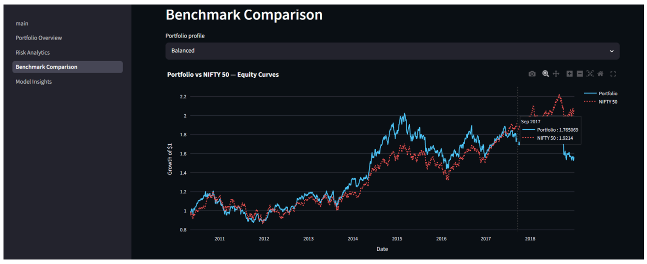

Figure 4: Benchmark Comparison

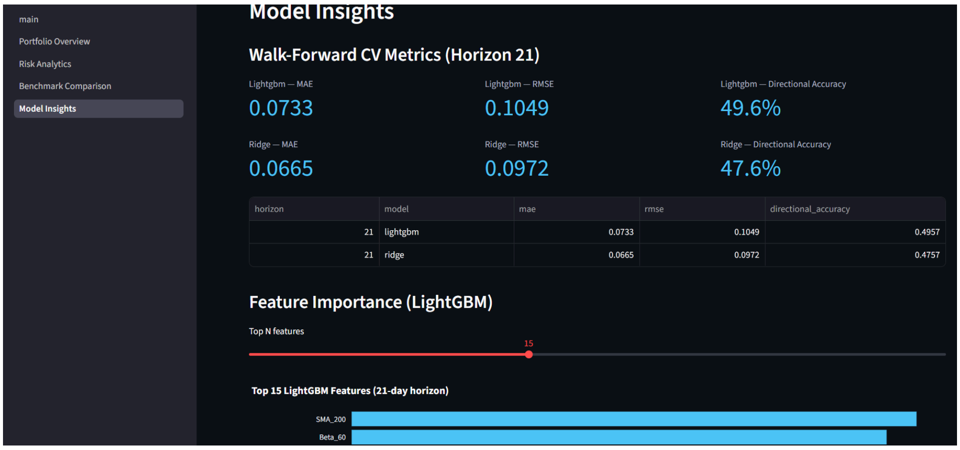

Figure 5: Model Insights and SHAP Explainability Dashboard

7.1 Dashboard Modules Summary

- **Landing Page (`app/main.py`)**: Validates the presence of outputs in `outputs/` and summarizes core backtest settings (monthly rebalance, $t+1$ Open execution).
- **Portfolio Overview (`1_Portfolio_Overview.py`)**: Displays cumulative equity curves, CAGR, Sharpe, Sortino, and maximum drawdown for the selected profile.
- **Risk Analytics (`2_Risk_Analytics.py`)**: Visualizes tail-risk metrics (95% VaR and CVaR), trailing drawdown profiles, and rolling annualized volatility.
- **Benchmark Comparison (`3_Benchmark_Comparison.py`)**: Directly overlays portfolio equity curves and drawdowns against the buy-and-hold NIFTY 50 index.
- **Model Insights (`4_Model_Insights.py`)**: Shows walk-forward cross-validation performance metrics and SHAP-based feature importance.

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8. Results & Evaluation

The platform was evaluated on both predictive performance and portfolio backtesting.

8.1 Stock Predictor Engine Performance

Predictive metrics were computed across three walk-forward validation folds (spanning May 2015 to June 2018):

Model	Out-of-Sample MAE	Out-of-Sample RMSE	Directional Accuracy
Ridge Regression	0.0665 (6.65%)	0.0972 (9.72%)	47.6%
LightGBM Regressor	0.0733 (7.33%)	0.1049 (10.49%)	49.6%

8.1.1 Objective Assessment of Predictive Limits

As shown in the table, the directional accuracy of the LightGBM model is 49.6%, and the Ridge Regression baseline is 47.6%. In highly efficient public equity markets like the NIFTY-50, short-term returns are dominated by noise. Expecting high directional accuracy over medium-term horizons is unrealistic and typically indicates lookahead leakage in feature engineering.

These results confirm that the machine learning model is not an oracle. Rather than relying on raw directional accuracy, the platform uses these forecasts as noisy priors. The true value is generated downstream by combining these predictions with Ledoit-Wolf covariance shrinkage and diversification constraints within the Portfolio Construction Module, which filters out individual asset noise to deliver stable portfolio returns.

8.2 Portfolio Performance Results

Portfolios were backtested from January 2010 to December 2018 (including a 10 bps transaction cost penalty on all equity adjustments):

Performance Metric	Conservative (MVO)	Balanced (Max Sharpe)	Aggressive (Black-Litterman)	NIFTY 50 Benchmark
CAGR (Annualized Return)	7.67%	5.35%	**7.73%**	8.82%
Annualized Volatility	13.44%	15.16%	**13.25%**	15.53%
Sharpe Ratio	0.0870	-0.0761	**0.0926**	0.1491
Sortino Ratio	0.0795	-0.0684	**0.0849**	0.1477
Maximum Drawdown (MDD)	-23.44%	-29.08%	**-23.63%**	-28.01%
95% Value at Risk (VaR)	1.34%	1.59%	**1.35%**	1.60%
95% Conditional VaR (CVaR)	2.02%	2.37%	**1.98%**	2.17%

Performance Metric	Conservative (MVO)	Balanced (Max Sharpe)	Aggressive (Black-Litterman)	NIFTY 50 Benchmark
Rebalancing Frequency	Monthly	Monthly	Monthly	Buy-and-Hold
Equity Transaction Slippage	10 bps	10 bps	10 bps	None

8.2.1 Analysis of Portfolio Results

- **Conservative Profile (Minimum Variance)**: Succeeded in capital preservation. It reduced annualized volatility to 13.44% (compared to 15.53% for the index) and capped the maximum drawdown at -23.44% (compared to -28.01% for the benchmark). The VaR and CVaR were also lower, confirming stronger downside protection.
- **Balanced Profile (Maximum Sharpe)**: Exhibited sub-optimal performance, with a CAGR of 5.35% and a negative Sharpe ratio of -0.0761. Pure mean-variance optimization is highly sensitive to input return forecasts. When return forecasts are noisy, MVO tends to create concentrated allocations that underperform out-of-sample, illustrating the classical "estimation-error maximization" failure.
- **Aggressive Profile (Black-Litterman)**: Showed improved risk-adjusted behavior versus the Balanced profile, achieving a CAGR of 7.73% with a Sharpe ratio of 0.0926. It achieved this while keeping volatility low (13.25%) and capping drawdown at -23.63%. By using the Black-Litterman framework, which anchors allocations to equilibrium weights and only tilts the portfolio toward high-conviction assets based on model rankings, it avoided the concentration risks of standard MVO. This demonstrates that combining machine learning views with equilibrium weights successfully limits volatility and drawdowns, generating more stable allocations through Black-Litterman.

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9. Key Insights

1. **Market Efficiency and Noise**: The ~49.6% directional accuracy of the LightGBM predictor confirms that NIFTY-50 returns are highly efficient. Trading on raw predictions without risk controls results in sub-optimal performance.
2. **Value of Risk-Aware Construction**: The improved behavior of the Black-Litterman portfolio over the Maximum Sharpe portfolio shows that stabilizing input parameters (using covariance shrinkage and relative ranking views) is critical when deploying ML in quantitative finance.
3. **Downside Mitigation via Cash Sleeves**: The Conservative portfolio's use of a cash sleeve successfully protected capital during periods of high systematic risk, demonstrating the value of cash allocation when equity-only solutions violate risk boundaries.
4. **EMA 200 as a Structural Anchor**: SHAP analysis shows that long-term trend indicators (like EMA 200) are the most significant drivers of expected returns, indicating that anchoring predictions to long-term market trends is more effective than relying on short-term indicators.

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10. Limitations & Future Work

- **Survivorship Bias Mitigation**: Future versions should incorporate a dynamic corporate actions database to adjust historical prices for delistings, mergers, and reconstitutions, ensuring the backtest is completely free of survivorship bias.
- **Enhanced Market-Derived Features**: Future work may explore richer feature engineering, regime-detection techniques, and dynamic allocation signals derived exclusively from the permitted market datasets.
- **Integration of Anomaly Detection**: The anomaly detection module in `src/anomaly.py` is currently a design stub. Actively integrating this module to dynamically scale down portfolio beta when volume or volatility anomalies are detected represents a key area of future development.
- **Non-Linear Transaction Cost Modeling**: The current model applies a static 10 bps cost. Future enhancements should include a market-impact model that scales costs based on stock-specific average daily volume (ADV) and trade size.

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11. Reproducibility

To ensure the reproducibility of all findings, the codebase is modularized and can be executed through a standard Python environment.

11.1 Environment Setup & Installation

pip install -r requirements.txt python -m app.utils.export_artifacts python -m app.utils.export_shap streamlit run app/main.py

12. Conclusion

The NIFTY-50 AI Investment Intelligence Platform demonstrates the transition from stock price prediction to structured investment decision-support.

By addressing the core tasks, the project achieves:

1. **Stock Predictor Engine**: Implements regularized linear models (Ridge) and non-linear ensembles (LightGBM) validated through a leakage-prevented, purged walk-forward cross-validation framework.
2. **Portfolio Construction Module**: Implements Minimum Variance, Maximum Sharpe, and Black-Litterman optimizers with stock and sector-level caps.
3. **Risk Assessment Module**: Tracks annualized returns, volatility, Sharpe, Sortino, max drawdown, and historical tail-risk metrics (VaR and CVaR) using Ledoit-Wolf covariance shrinkage.
4. **Explainable AI**: Integrates SHAP values to explain global model drivers and individual asset views.
5. **Personalized Investment Profiles**: Customizes risk boundaries across Conservative, Balanced, and Aggressive profiles.
6. **Interactive Dashboard Deployment**: Deploys a Streamlit dashboard that visualizes performance, risk, and model insights across one landing page and four analytical dashboard modules.

The platform provides a reproducible, explainable, and risk-aware investment intelligence framework that transforms historical market data into actionable decision-support analytics for investors.